

Activity - 5

① Given 4371, 1323, 6173, 4199, 4344,
9678, 1989

i) Separate Chaining

$$4371 \% 10 \rightarrow 1$$

$$1323 \% 10 \rightarrow 3$$

$$6173 \% 10 \rightarrow 3$$

$$4199 \% 10 \rightarrow 9$$

$$4344 \% 10 \rightarrow 4$$

$$9679 \% 10 \rightarrow 9$$

$$1989 \% 10 \rightarrow 9$$

0	
1	→ 4371
2	
3	→ 1323 → 6173
4	→ 4344
5	
6	
7	
8	
9	→ 4199 → 9679 → 1989

ii) Linear Probing

$$h(\text{key}) = h(\text{key} + i)$$

9679	0
4371	1
1989	2
1323	3
6173	4
4344	5
	6
	7
	8
4199	9

ii) Quadratic Probing

9679	0
4371	1
	2
1323	3
6173	4
<u>4344</u>	5
	6
	7
1989	8
4199	

iii) Double Hashing

0	
1	4371
2	
3	1323
4	6173
5	9679
6	
7	4344
8	1989
9	4199

$$9 + (7 - (1989 \% 7)) \times 1$$

$$9 + (7 - 1) \times 1$$

$$9 + 6$$

$$15$$

$$2 - 9 + 12 = 21$$

$$3 - 9 + 18 = 27$$

$$4 - 9 + 24 = 33$$

$$5 - 9 + 30 = 39$$

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